

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

$$= \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$$\alpha, \beta \in \mathbb{C}$$

let's represent $|\psi\rangle$ in polar form

$$|\psi\rangle = a e^{i\theta} |0\rangle + b e^{i\varphi} |1\rangle$$

$$a, b \in \mathbb{R}$$

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$$= e^{i\theta} \left(a|0\rangle + e^{i(\varphi-\theta)} |1\rangle \right)$$

global phase
does not change probabilities
can be ignored.

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global phase
does not change probabilities
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$$= a|0\rangle + be^{i\varphi}|1\rangle$$

Arbitrary quantum
state of a qubit

$$|\psi\rangle = a|0\rangle + be^{i\phi}|1\rangle$$

$$a^2 + b^2 = 1$$

Recall the state $|\psi\rangle = \frac{3}{5}|0\rangle + \frac{4}{5}e^{i\pi/6}|1\rangle$

Arbitrary quantum
state of a qubit

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$$a^2 + b^2 = 1$$

Recall the state $|\psi\rangle = \frac{3}{5}|0\rangle + \frac{4}{5}e^{i\pi/6}|1\rangle$

since $a^2 + b^2 = 1$ let's apply a trigonometric transformation

$$a = \cos\left(\frac{\theta}{2}\right) \quad b = \sin\left(\frac{\theta}{2}\right)$$

Arbitrary quantum
state of a qubit

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + \sin\left(\frac{\theta}{2}\right)e^{i\phi}|1\rangle$$

Represent any $|\psi\rangle$ w/ just $\{\theta, \phi\}$

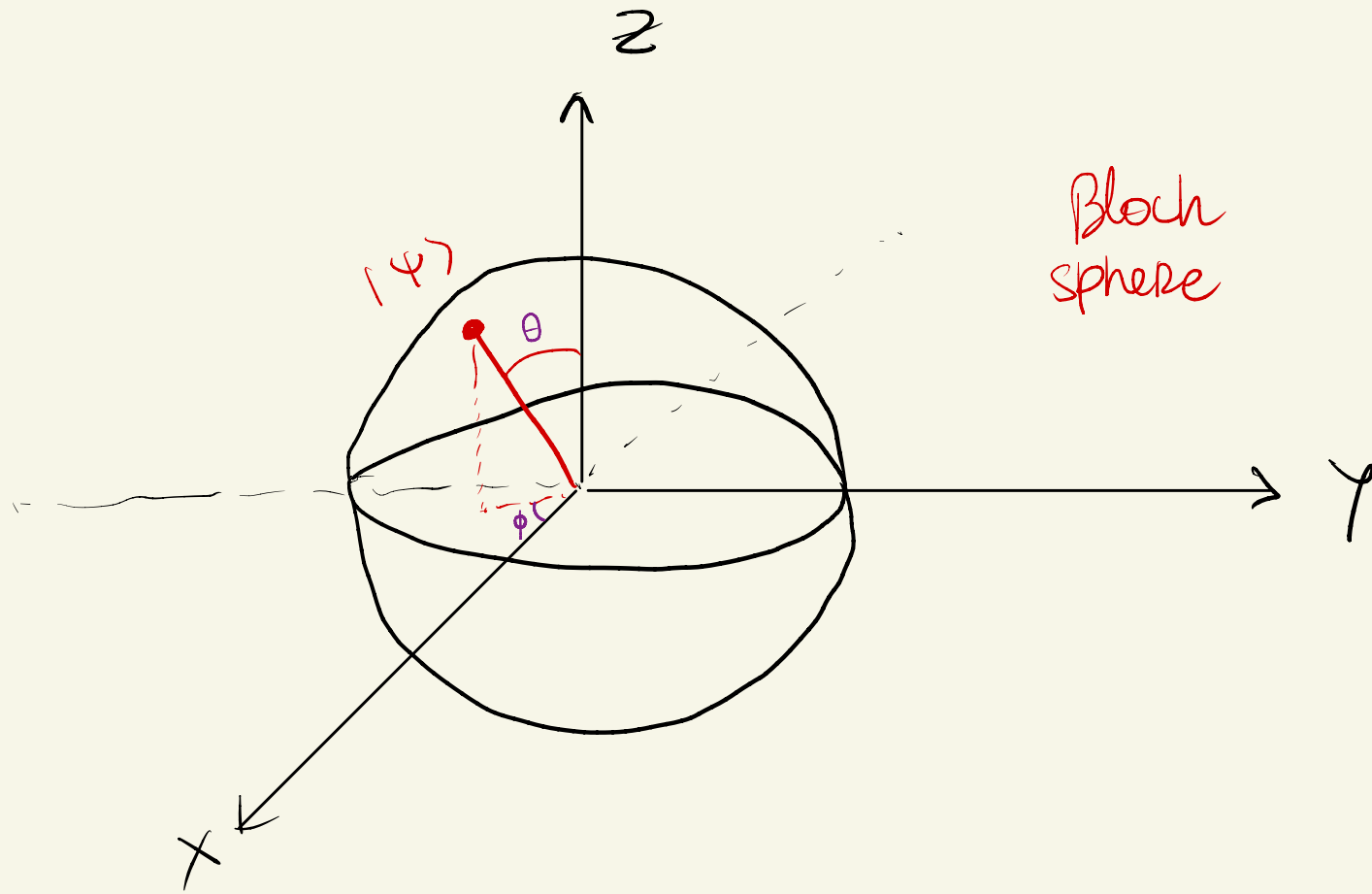
(But why are we doing this?)

Arbitrary quantum
Arbitrary quantum
state of a qubit
state of a qubit

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + \sin\left(\frac{\theta}{2}\right)e^{i\phi}|1\rangle$$

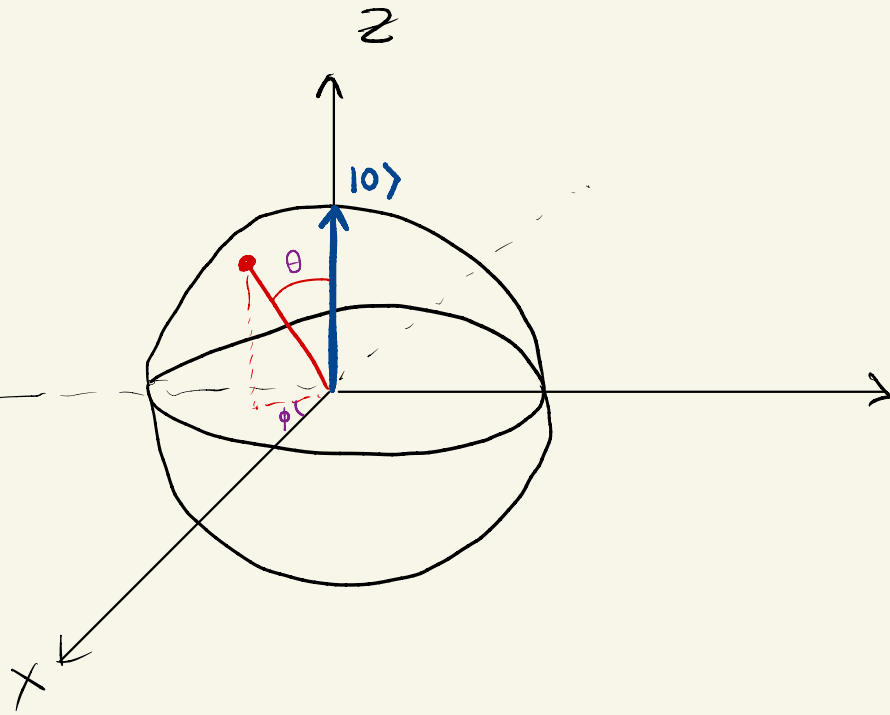
Represent any $|\psi\rangle$ w/ just $\{\theta, \phi\}$

Natural interpretation in spherical coordinates

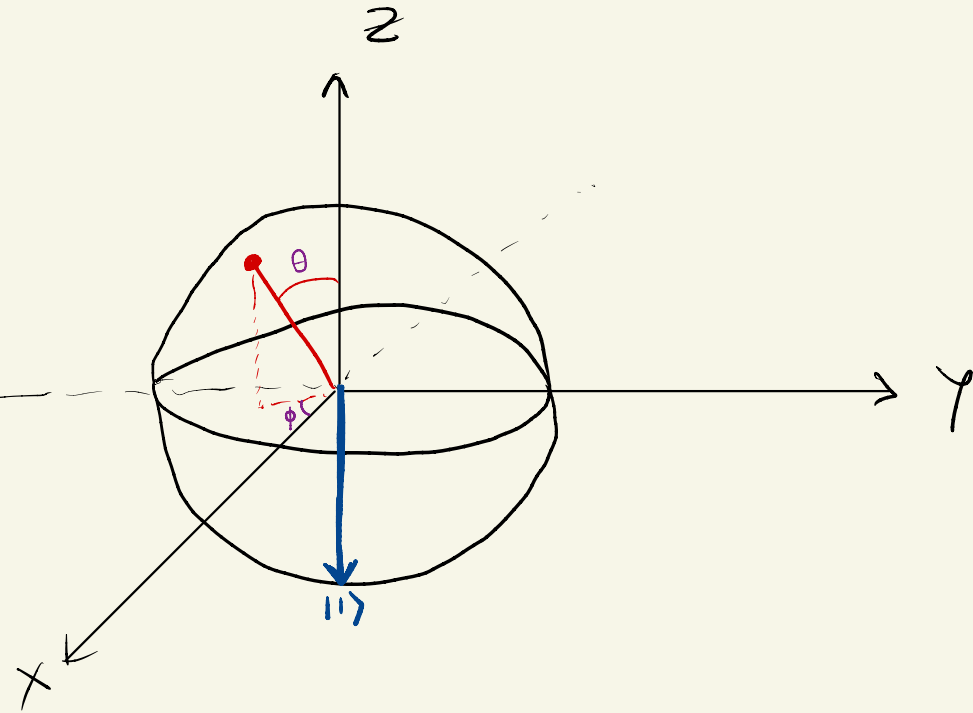


classical bits : deterministic states

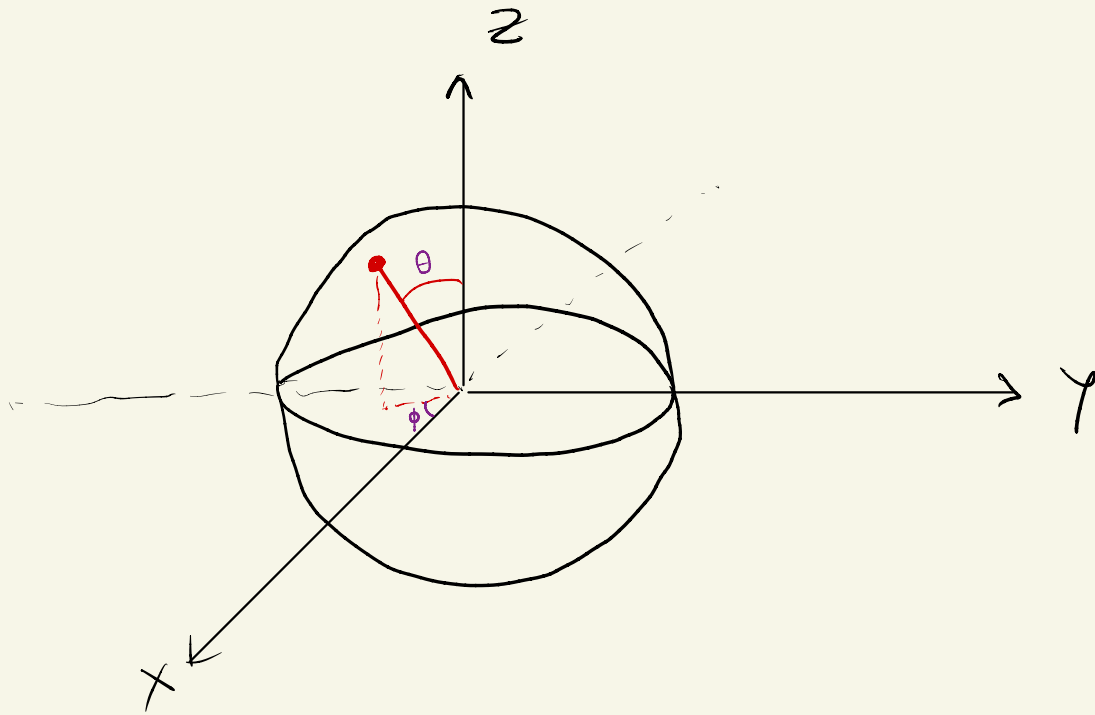
$$\theta = \phi = 0$$
$$|\psi\rangle = |0\rangle$$



$$\theta = \pi$$
$$\phi = 0$$
$$|\psi\rangle = |1\rangle$$

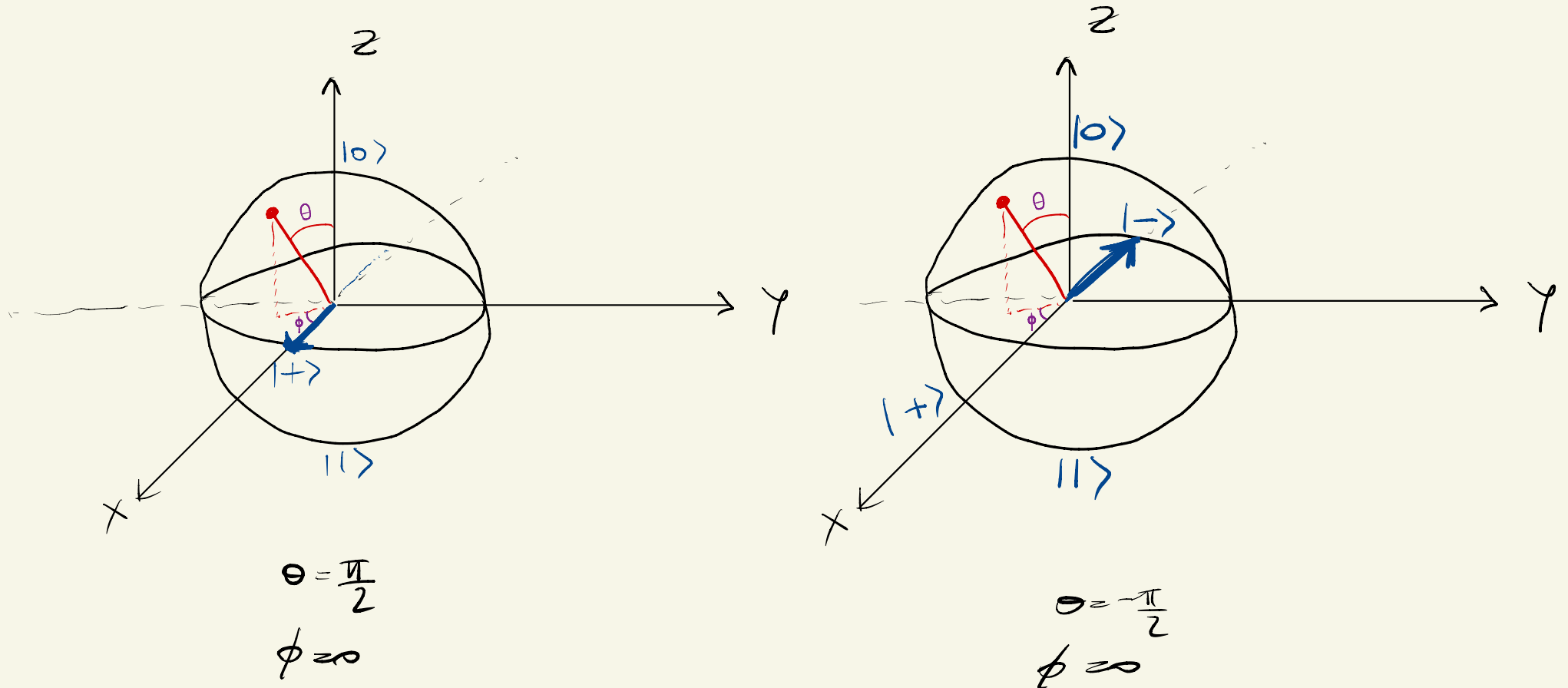


Quantum bits can occupy a multitude
(infinitely many) of states

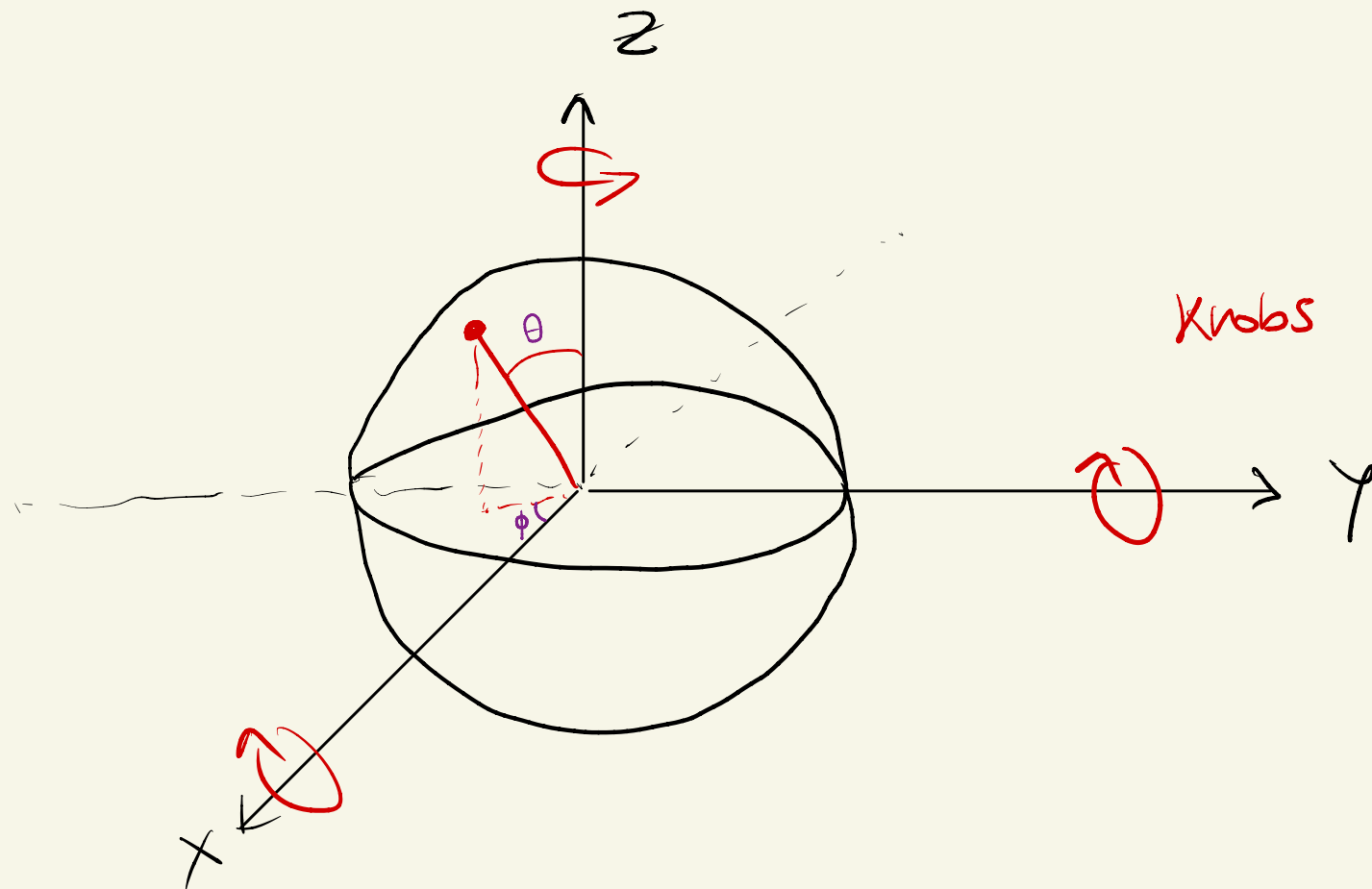


unitary superposition
states ?

Quantum bits can occupy a multitude
(infinitely many) of states

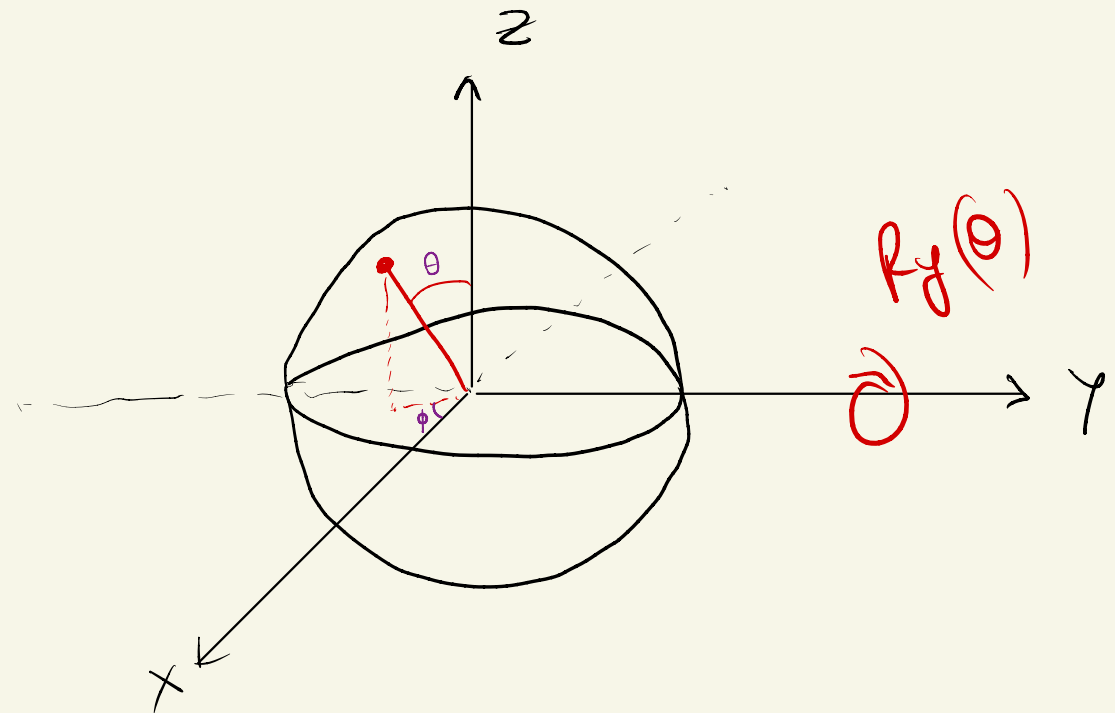


How can we change the amplitudes of quantum states?



Recall

$$R_y(\theta) = \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) & -\sin\left(\frac{\theta}{2}\right) \\ \sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{pmatrix}$$



Rotation gates

$$R_x(\theta) = \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) & -i\sin\left(\frac{\theta}{2}\right) \\ i\sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{pmatrix}$$

$$R_z(\theta) = \begin{pmatrix} e^{-i\theta} & 0 \\ 0 & e^{i\theta} \end{pmatrix}$$

Gate set so far

$$\{H, X, Y, Z, P_\phi, R_Y, R_X, R_Z\}$$

2 qubit operations

$$\text{ex: } |0\rangle \otimes |0\rangle$$



tensor product

$$|\psi\rangle = \begin{pmatrix} \alpha_0 \\ \beta_0 \end{pmatrix}$$

$$|\phi\rangle = \begin{pmatrix} \alpha_1 \\ \beta_1 \end{pmatrix}$$

$$|\psi\rangle \otimes |\phi\rangle = \begin{pmatrix} \alpha_0 \\ \beta_0 \end{pmatrix} \otimes \begin{pmatrix} \alpha_1 \\ \beta_1 \end{pmatrix}$$

$$= \begin{pmatrix} \alpha_0 \begin{pmatrix} \alpha_1 \\ \beta_1 \end{pmatrix} \\ \beta_0 \begin{pmatrix} \alpha_1 \\ \beta_1 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} \alpha_0 \alpha_1 \\ \alpha_0 \beta_1 \\ \beta_0 \alpha_1 \\ \beta_0 \beta_1 \end{pmatrix}$$

Ex : Represent all two qubit
basis states

$|00\rangle$

$|01\rangle$

$|10\rangle$

$|11\rangle^2$

Ex : Represent all two qubit
basis states

$$|00\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$|01\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

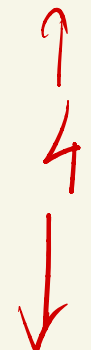
$$|10\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

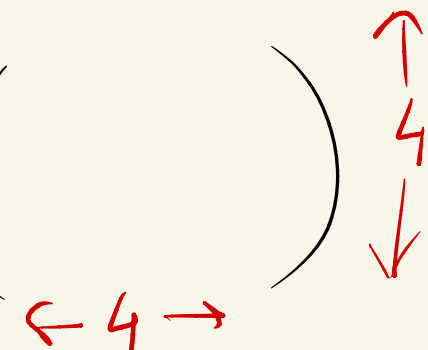
$$|11\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

So, for two qubit states $|\psi\rangle$

$$|\psi'\rangle = U |\psi\rangle$$

where $|\psi\rangle = \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}$



$$U = \begin{pmatrix} & & & \\ & & & \\ & & & \\ & & & \end{pmatrix}$$


Sometimes,

$$U \cong U_{2 \times 2} \otimes U_{2 \times 2}$$

E.g. $U_a = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$$U_a \otimes U_b = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \otimes \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$$

$$U_b = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$$

$$= \begin{pmatrix} a \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} & b \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \\ c \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} & d \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \end{pmatrix}$$

$$(U_a \otimes U_b) (|\psi\rangle \otimes |\phi\rangle)$$

$$= U_a |\psi\rangle \otimes U_b |\phi\rangle$$

Ex: $\left. \begin{array}{l} |0\rangle - \boxed{H} - \\ |0\rangle - \boxed{H} - \end{array} \right\}$ Resulting quantum state?

Either calculate tensor product

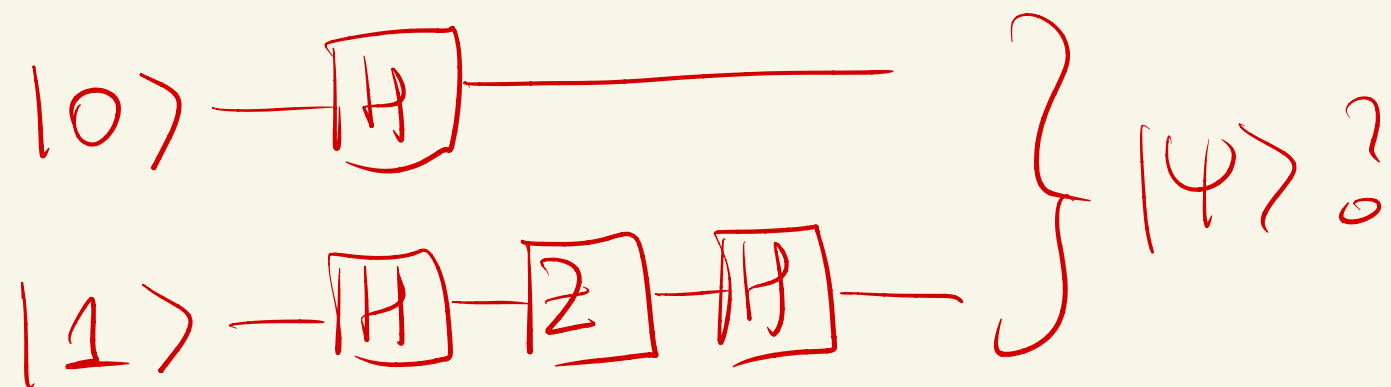
$$H \otimes H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \otimes \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

and vector $|00\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$

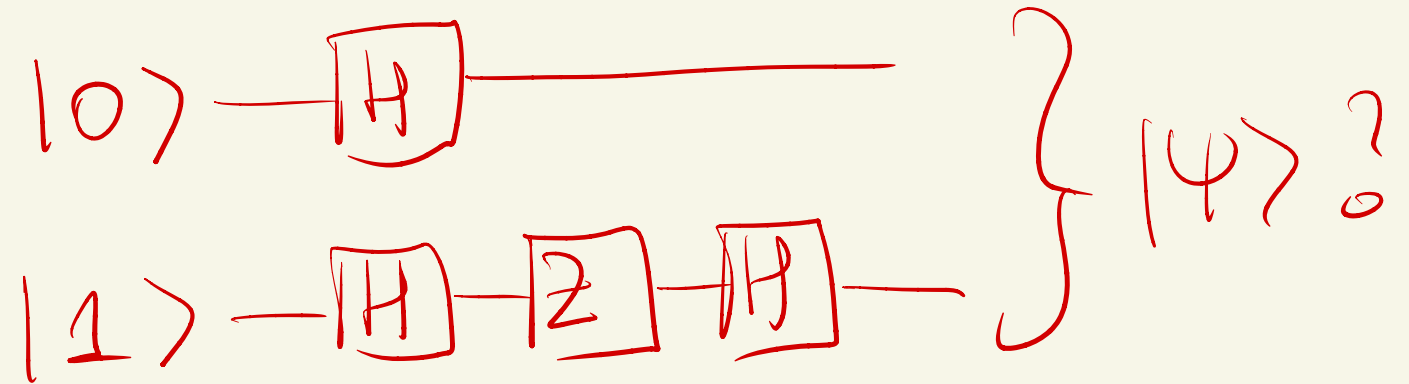
or Dirac notation:

$$\begin{aligned} (H \otimes H) |0\rangle \otimes |0\rangle &= \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \\ &= \frac{1}{2} |00\rangle + \frac{1}{2} |01\rangle + \frac{1}{2} |10\rangle + \frac{1}{2} |11\rangle \end{aligned}$$

Ex 1



Ex 1



$$H|0\rangle \otimes X|1\rangle$$

$$= \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle$$

$$= \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|10\rangle$$

Ex: $|\psi\rangle = \frac{3}{5\sqrt{2}}|00\rangle - \frac{3}{5\sqrt{2}}|01\rangle - \frac{4}{5\sqrt{2}}e^{i\pi/6}|10\rangle$
 $+ \frac{4}{5\sqrt{2}}e^{i\pi/6}|11\rangle$

circuit that prepares $|\psi\rangle$?

$$|0\rangle \text{---} \boxed{\text{Ry}(\theta)} \text{---} \boxed{\text{P}\phi}$$

$$|1\rangle \text{---} \boxed{\text{H}} \text{---}$$

$$\theta = 2\arccos\left(\frac{3}{5}\right)$$

$$\phi = \pi - \theta/6$$

why?!

$$|0\rangle \xrightarrow{R_y(\theta)} |P_\theta\rangle$$

$$|1\rangle \xrightarrow{H}$$

$$\theta = \arccos\left(\frac{3}{5}\right)$$

$$\phi = \pi - \pi/6$$

$$\frac{3}{5\sqrt{2}} |00\rangle - \frac{3}{5\sqrt{2}} |01\rangle$$

$$- \frac{4}{5\sqrt{2}} e^{i\pi/6} |10\rangle + \frac{4}{5\sqrt{2}} e^{i\pi/6} |11\rangle$$

Recall the state from last lecture:

$$|\phi\rangle = \frac{3}{5} |0\rangle - \frac{4}{5} e^{i\pi/6} |1\rangle$$

what if we do $|\phi\rangle \otimes \begin{pmatrix} \alpha_0 \\ \beta_0 \end{pmatrix}$?

$$|\phi\rangle \otimes \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \left(\frac{3}{5}|0\rangle - \frac{4}{5}e^{i\pi/6}|1\rangle \right) \otimes (\alpha_0|0\rangle + \beta_0|1\rangle)$$

$$= \frac{3}{5}\alpha_0|00\rangle + \frac{3}{5}\alpha_1|01\rangle$$

$$- \frac{4}{5}e^{i\pi/6}\alpha_0|10\rangle - \frac{4}{5}e^{i\pi/6}\alpha_1|11\rangle$$

What should be the value
of α_0, β_0 ?

$$|\phi\rangle \otimes \begin{pmatrix} \alpha \\ \beta_0 \end{pmatrix} = \left(\frac{3}{5}|0\rangle - \frac{4}{5}e^{i\pi/6}|1\rangle \right) \otimes (\alpha_0|0\rangle + \beta_0|1\rangle)$$

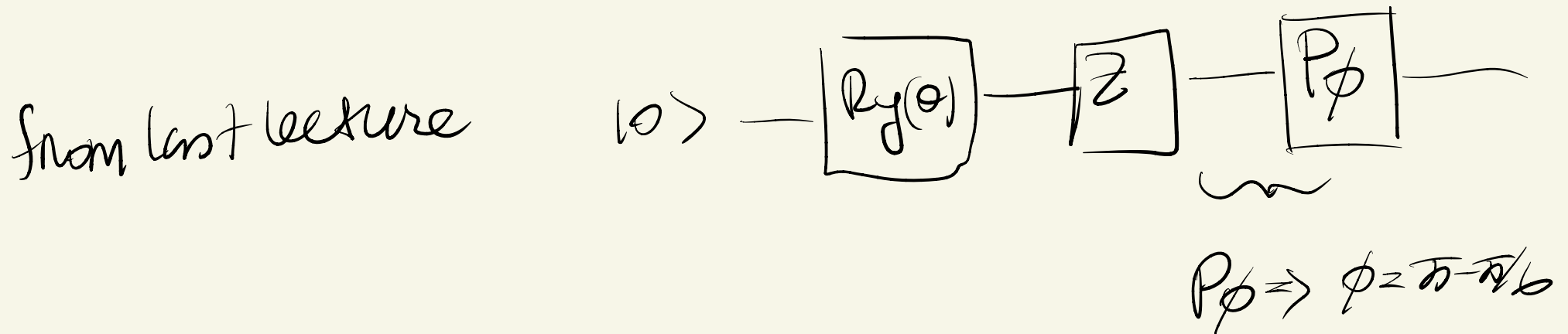
$$= \frac{3}{5}\alpha_0|00\rangle + \frac{3}{5}\alpha_1|01\rangle$$

$$- \frac{4}{5}e^{i\pi/6}\alpha_0|10\rangle - \frac{4}{5}e^{i\pi/6}\alpha_1|11\rangle$$

if $\alpha_0 = \frac{1}{\sqrt{2}}$ and $\alpha_1 = -\frac{1}{\sqrt{2}}$ we get $|\psi\rangle$

$$\text{So } \begin{pmatrix} \alpha_0 \\ \beta_0 \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix} = |-\rangle = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$$

How to prepare $|\phi\rangle = \frac{3}{5}|0\rangle - \frac{4}{5}e^{i\pi/6}|1\rangle$?



$$\cos\left(\frac{\theta}{2}\right) = \frac{3}{5} \Rightarrow \theta = 2\arccos\left(\frac{3}{5}\right)$$

Finally, the circuit that prepares $|\Psi\rangle$

